6 Fig. 6.1 shows an object O which is 5.0 cm away from the centre of a thin, converging lens L. The focal length of L is 3.0 cm. Fig. 6.1 is drawn to full scale.

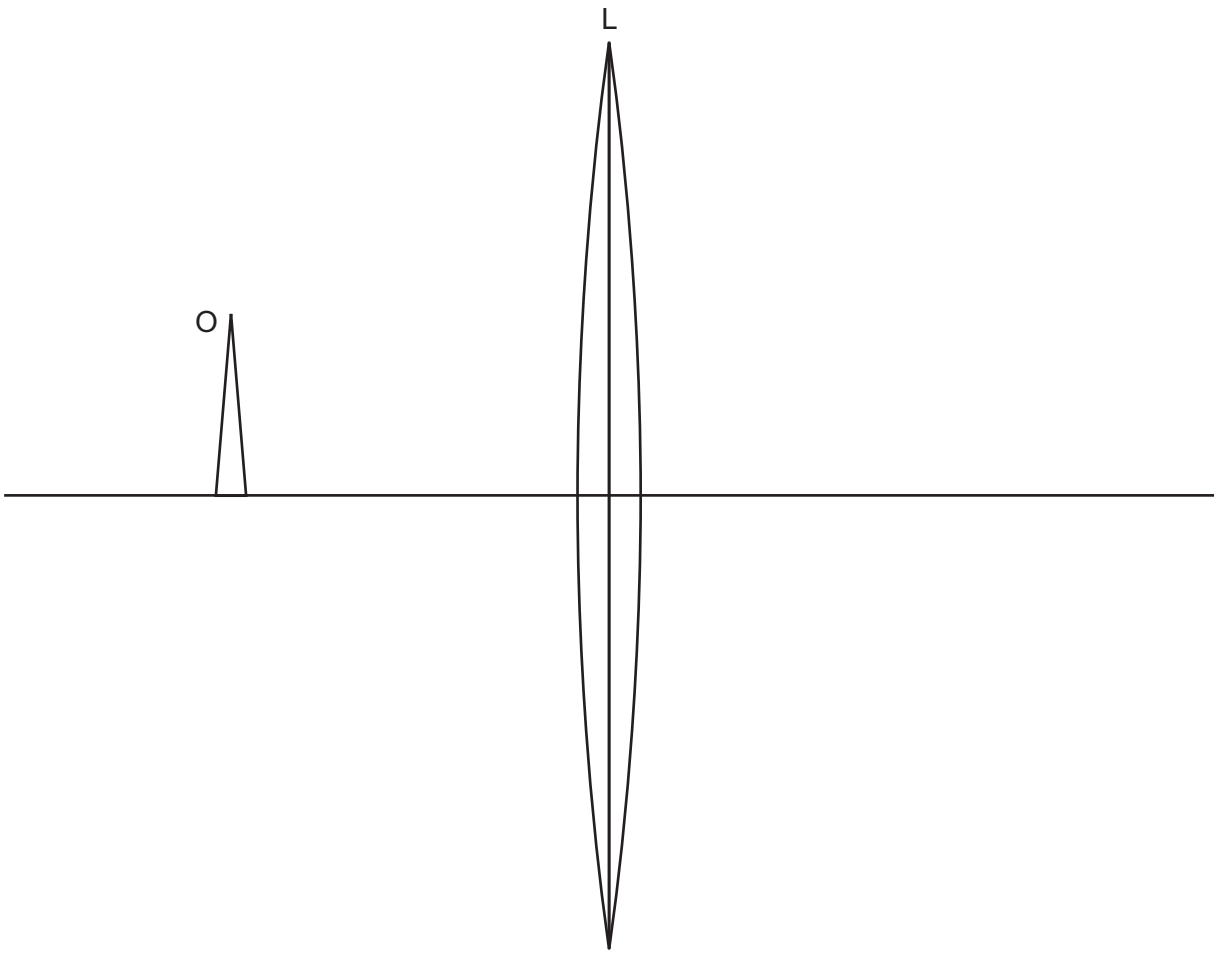

Fig. 6.1

(a) (i) On Fig. 6.1, label the principal axis with a P. [1]

(ii) On Fig. 6.1, place a letter X at a focal point. [1]

(iii) On Fig. 6.1, draw two rays from O to locate the tip of the image produced by the lens. [2]

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(iv) In Table 6.1, place a tick in the right-hand column next to **all** the terms that describe the image in **(a)(iii)**.

Table 6.1

diminished	
enlarged	
inverted	
real	
same size	
upright	
virtual	

[3]

(b) The object moves closer to L. The new distance between L and the object is less than the focal length of L.

Describe how the new image is different from the image in **(a)(iv)**.

.....

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..... [2]

[Total: 9]

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